

Problem 1

$$\begin{aligned} P F \quad \text{of} \quad 12 &= 2 \times 6 \\ &= 2 \times 2 \times 3 \\ &= 2^2 \times 3' \end{aligned}$$

$$\begin{aligned} P F \quad \text{of} \quad 18 &= 2 \times 9 \\ &= 2 \times 3 \times 3 \\ &= 2' \times 3^2 \end{aligned}$$

$$\begin{aligned} G C D (12, 18) &= 2' \times 3' \\ &= 2 \times 3 \\ &= 6 \end{aligned}$$

$$N O. \quad \text{of} \quad 2' s \quad \text{powers} = 1 + 1$$

$$= 2$$

$$N O. \quad \text{of} \quad 3' s \quad \text{power} s = 1 + 1$$

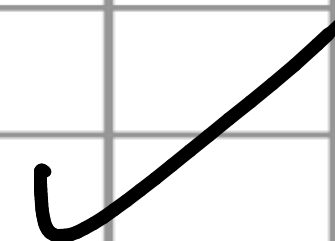
$$= 2$$

$$T o t a l \quad f a c t o r s = 2 \times 2$$

$$= (1+1) (1+1)^4 = 4$$

It's confusing

List of factors = $[(2^0 \times 3^0), (2^1 \times 3^0), (2^0 \times 3^1), (2^1 \times 3^1)]$
= $[1, 2, 3, 6]$



problem 2:

P F 0 f 2 4 $=$ 8 $\times$ 3
 $=$ 2 $\times$ 2 $\times$ 3

P F 0 f 3 6 $=$ 2^3 $\times$ 3^1
 G C D $($ 2 4 $,$ 3 6 $)$ $=$ 2^2 $\times$ 3^2
 $=$ 4 $\times$ 3
 $=$ 1 2

N O $.$ 0 f 2 $'$ s p o w e r s $=$ 2 $+$ 1

N O $.$ 0 f 3 $'$ s p o w e r s $=$ 3 $+$ 1

T O t a l n o $.$ 0 f f a c t o r s $=$ 3 $\times$ 2

L i s t o f f a c t o r s $=$ $[$ $($ 2^0 $\times$ 3^0 $)$ $,$ $($ 2^1 $\times$ 3^0 $)$ $,$ $($ 2^0 $\times$ 3^1 $)$ $,$ $($ 2^1 $\times$ 3^1 $)$ $]$
 $=$ $[$ $1, 2, 4, 3, 6, 12]$

Problem 3:

$$\begin{aligned} P F \quad \text{of} \quad 40 &= 8 \times 5 \\ &= 2 \times 2 \times 2 \times 5 \\ &= 2^3 \times 5^1 \end{aligned}$$

$$\begin{aligned} P F \quad \text{of} \quad 60 &= 1 \times 2 \times 5 \\ &= 2 \times 2 \times 3 \times 5 \\ &= 2^2 \times 3^1 \times 5^1 \end{aligned}$$

$$\begin{aligned} G C D &= 2^2 \times 3^0 \times 5^1 \\ &= 4 \times 5 \\ &= 20 \end{aligned}$$

$$\begin{aligned} \text{Ans} &= (2+1)(1+1) \\ &= 3 \times 2 = 6 \end{aligned}$$

$$\begin{aligned} N O . \quad \text{of} \quad 2's \text{ powers} &= 2 + 1 \\ &= 3 \end{aligned}$$

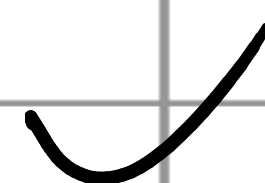
$$\begin{aligned} N O . \quad \text{of} \quad 3's \text{ powers} &= 0 + 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} N O . \quad \text{of} \quad 5's \text{ powers} &= 1 + 1 \\ &= 2 \end{aligned}$$

$$\text{Total factors} = 3 \times 1 \times 2 \\ = 6$$

List of factors $\rightarrow$

$$\rightarrow = [(2^0 \times 3^0 \times 5^0), (2^1 \times 3^0 \times 5^0), (2^2 \times 3^0 \times 5^0), \\ (2^0 \times 3^0 \times 5^1), (2^1 \times 3^0 \times 5^1), (2^2 \times 3^0 \times 5^1)] \\ = [1, 2, 4, 5, 10, 20]$$



Problem 4:

$$P F \quad 0 f \quad 1 5 = 5 \times 3$$

$$= 5' \times 3'$$

$$P F \quad 0 f \quad 7 5 = 2 5 \times 3$$

$$= 5 \times 5 \times 3$$

$$= 5^2 \times 3'$$

$$G C D = 5' \times 3'$$

$$= 1 5$$

$$N O. \quad 0 f \quad 5' s \quad p o w e r s = 1 + 1$$

$$= 2$$

$$N O. \quad 0 f \quad 3' s \quad p o w e r s = 1 + 1$$

$$= 2$$

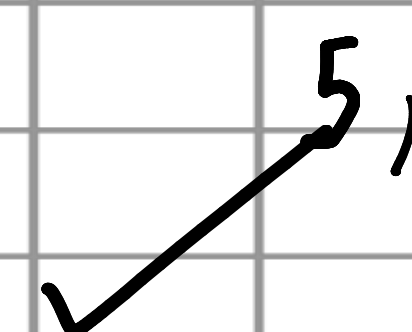
$$2 \times 2 = 4$$

$$L i s t \quad o f \quad f a c t o r s = [(5^0 \times 3^0), (5' \times 3^0),$$

$$(5^0 \times 3^1), (5' \times 3^1)]$$

$$= [1, 5, 3,$$

$$1 5]$$



Problem 5:

$$P F (48) = 16 \times 3$$

$$= 2 \times 2 \times 2 \times 2 \times 3$$

$$= 2^4 \times 3^1$$

$$P F (80) = 8 \times 10$$

$$= 2 \times 2 \times 2 \times 2 \times 5$$

$$= 2^4 \times 5^1$$

$$G C D = 2^4 \times 3^0 \times 5^0$$

$$= 16$$

$$\text{No. of } 2\text{'s power} = 4 + 1$$

$$= 5$$

$$\text{No. of } 3\text{'s power} = 0 + 1$$

$$= 1$$

$$\text{No. of } 5\text{'s power} = 0 + 1$$

$$= 1$$

$$5 \times 1 \times 1 = 5$$

Not needed

Not needed

List of factors = $(2^0 \times 3^0 \times 5^0), (2^1 \times 3^0 \times 5^0), (2^2 \times 3^0 \times 5^0), (2^3 \times 3^0 \times 5^0), (2^4 \times 3^0 \times 5^0)$

$= [0, 2, 4, 8, 16]$

1

Problem 6:

$$P F (1 \ 0 \ 0) = 4 \times 2 \ 5$$

$$= 2 \times 2 \times 5 \times 5$$

$$= 2^2 \times 5^2$$

$$P F (1 \ 5 \ 0) = 5 \ 0 \times 3$$

$$= 2 \times 5 \times 5 \times 3$$

$$= 2^1 \times 5^2 \times 3^1$$

$$G C D = 2^1 \times 5^2 \times 3^0$$

$$= 50$$

$$\text{No. of } 2^1 \leq \text{powers} = 1 + 1$$

$$= 2$$

$$\text{No. of } 5^1 \leq \text{powers} = 2 + 1$$

$$= 3$$

$$\text{No. of } 3^1 \leq \text{powers} = 0 + 1$$

$$= 1$$

$$2 \times 3 \times 1 = 6 \quad \checkmark$$

List of factors = $[(5^0 \times 2^0 \times 3^0, (5^1 \times 2^0 \times 3^0, (5^2 \times 2^0 \times 3^0, (5^1 \times 2^1 \times 3^0, (5^2 \times 2^1 \times 3^0)]$
 $(5^0 \times 2^1)$
 $= [1, 5, 2, 5, 1, 0, 5, 0]$

2
2

Problem 7:

$$P F (5 \ 6) = 8 \times 7$$

$$= 2 \times 2 \times 2 \times 7$$

$$= 2^3 \times 7^1$$

$$P F (9 \ 8) = 1 \ 4 \times 7$$

$$= 2 \times 7 \times 7$$

$$= 2^1 \times 7^2$$

$$G C D = 2^1 \times 7^1$$

$$N o. \quad o f \quad 2's \quad f a c t o r s = 1 + 1$$

$$= 2$$

$$N o. \quad o f \quad 7's \quad f a c t o r s = 1 + 1$$

$$= 2$$

$$2 \times 2 = 4$$

$$L i s t \quad o f \quad f a c t o r s = [(2^0 \times 7^0), (2^1 \times 7^0),$$

$$(2^0 \times 7^1), (2^1 \times 7^1)]$$

$$= [1, 2, 7, 14]$$

Problem 8:

$$P F (8 \ 1) = 3 \times 2 \ 7$$

$$= 3 \times 3 \times 3 \times 3$$

$$= 3^4$$

$$P F (2 \ 7) = 3 \times 9$$

$$= 3 \times 3 \times 3$$

$$= 3^3$$

$$G C D = 3^3$$

$$= 2 \ 7$$

$$N O. \quad 0 \ A \quad 3' \ S \quad p o w e r \ S = 3 + 1$$

$$= 4$$

$$L i s t \quad o f \quad f a c t o r s = [3^0, 3^1, 3^2, 3^3]$$

$$= [1, 3, 9, 27]$$



Problem 9:

$$P F (4 \ 5) = 5 \times 9$$

$$= 5 \times 3 \times 3$$

$$= 5' \times 3^2$$

$$P F (1 \ 0 \ 5) = 5 \times 2 \ 1$$

$$= 5 \times 7 \times 3$$

$$= 5' \times 7' \times 3'$$

$$G_1 C D = 5' \times 7^0 \times 3'$$

$$= 1 \ 5$$

$$N O. \quad 0 \ f \quad 5' \ s \quad \text{factor} = 1 + 1$$

$$= 2$$

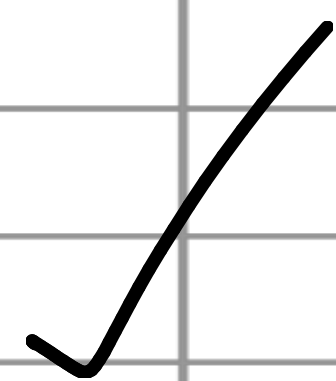
$$N O. \quad 0 \ f \quad 7' \ s \quad \text{factor} = 0 + 1$$

$$= 1$$

$$N O. \quad 0 \ f \quad 3' \ s \quad \text{factor} = 1 + 1$$

$$= 2$$

$$2 \times 1 \times 2 = 4$$



List of factors = $\{ (5^0 \times 7^0 \times 3^0),$
 $(5^1 \times 7^0 \times 3^0),$
 $(5^0 \times 7^0 \times 3^1),$
 $(5^1 \times 7^0 \times 3^1) \}$
 $= [1, 5, 3, 15]$



Problem 10:

$$P F (120) = 2 \times 60$$

$$= 5 \times 3 \times 2 \times 2 \times 2$$

$$= 5' \times 3' \times 2^3$$

$$P F (160) = 8 \times 20$$

$$= 2 \times 2 \times 2 \times 2 \times 2 \times 5$$

$$= 2^5 \times 5'$$

$$G C D = 2^3 \times 5' \times 3^0$$

$$= 40$$

$$\text{No. of } 2' \text{ s power s} = 3 + 1$$

$$= 4$$

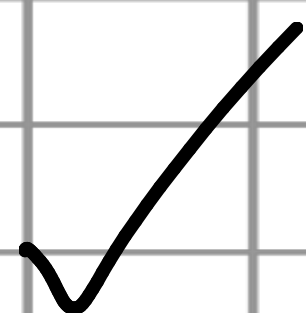
$$\text{No. of } 5' \text{ s power s} = 1 + 1$$

$$= 2$$

$$\text{No. of } 3' \text{ s power s} = 0 + 1$$

$$= 1$$

$$4 \times 2 \times 1 = 8$$



List of factors $\approx [(2^0 \times 5^0 \times 3^0),$
 $(2^1 \times 5^0 \times 3^0),$
 $(2^2 \times 5^0 \times 3^0),$
 $(2^3 \times 5^0 \times 3^0),$
 $(2^0 \times 5^1 \times 3^0),$
 $(2^1 \times 5^1 \times 3^0),$
 $(2^2 \times 5^1 \times 3^0),$
 $(2^3 \times 5^1 \times 3^0)]$

?

1, 2, 4, 5, 8, 10, 20, 40